

Cardiogenic shock as a complication of acute myocardial infarction: a nationwide cohort study

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Aims

Cardiogenic shock (CS) is a life-threatening complication of acute myocardial infarction (AMI). This study aimed to assess real-world incidence, characteristics, and outcomes in relation to age among patients with AMI complicated by CS (AMI-CS) compared with those without CS in a nationwide cohort.

Methods and results

Cohort study of all patients <85 years with AMI registered in the Norwegian Myocardial Infarction Registry 2013–2022. Outcomes were short- and long-term all-cause mortality. Mortality was assessed by Kaplan-Meier survival curves, Life Tables, and multivariable Cox regression models. Among 86 730 AMI patients (40% women), 1645 (2.7%) patients <75 years and 969 (3.9%) patients 75–84 years developed CS. Regardless of age, patients with AMI-CS were more likely to have previous heart failure, diabetes, renal failure, and ST-elevation AMI compared with patients without CS. In-hospital mortality was 47.5% in AMI-CS patients <75 years and 71.4% in patients 75–84 years, and the estimated one year cumulative mortality rates were 54.7% (95% CI: 52.3–57.1%) and 79.8% (95% CI: 77.2–82.3), respectively. Patients with AMI-CS who survived to hospital discharge remained at a higher long-term mortality risk compared with patients without CS (<75 years: adjusted HR 1.91 (95% CI 1.67–2.18), 75–84 years: adjusted HR 1.65 (95% CI: 1.41–1.93)).

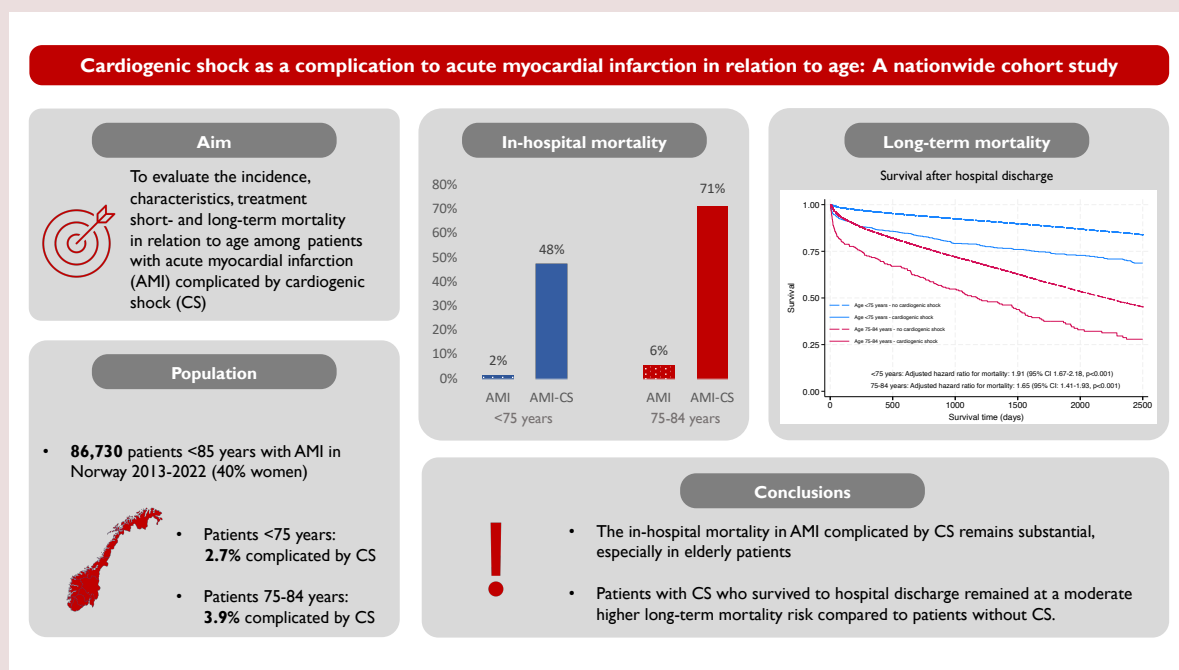
Conclusion

The in-hospital mortality rates for AMI-related CS remain high, especially in patients ≥75 years. However, long-term mortality in CS patients surviving to hospital discharge was only moderately higher compared with AMI patients without CS.

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Graphical Abstract



Keywords

Myocardial infarction • Cardiogenic shock • Incidence • Mortality

Introduction

Cardiogenic shock (CS) is a life-threatening complication of acute myocardial infarction (AMI).^{1,2} Together with cardiac arrest (CA), CS accounts for the majority of short-term mortality in AMI patients.¹⁻⁴ CS in the setting of AMI (AMI-CS) is characterized by acute cardiac dysfunction due to extensive ischaemia, leading to reduced cardiac output, hypotension and multi-organ failure.⁵ Immediate treatment of AMI-related CS includes urgent coronary revascularization, haemodynamic stabilization using vasoactive agents, respiratory support and, in some cases, temporary mechanical circulatory support (MCS).⁶ With the exception of revascularization, the evidence for other interventions is scarce. In the Should We Emergently Re-vascularize Occluded Coronaries for Cardiogenic Shock (SHOCK) trial, patients aged ≥ 75 years did not appear to have mortality benefit from early revascularization.⁷ However, this might be attributable to differences in baseline characteristics and limited sample size, and in recent years, also AMI-CS patients ≥ 75 years of age are regularly treated with early revascularization.^{8,9}

Inclusion of patients with AMI complicated by CS in randomized controlled trials is challenging due to a heterogeneous patient population, time-critical identification, inclusion and treatment, and ethical and communication barriers. Randomized trials represent a selected proportion of patients, elderly patients are rarely included, and there is a need for observational, real-world data on outcome in AMI-CS patients.

In recent decades, the incidence and mortality of AMI have decreased for all age groups, likely due to improvements in cardiovascular risk factors, increased use of acute reperfusion therapies, and better primary and secondary preventive treatment.^{5,10-12} Similar, a

decrease in the incidence of AMI related CS is described.¹³⁻¹⁵ However, it remains uncertain whether the mortality still is as high as previously described. Particularly, we need more information on the outcome of elderly patients with AMI-related CS. In addition to the worse outcome previously reported in this group of patients, initiating intensive care in elderly patients is resource consuming and may also raise ethical considerations.^{2,16}

The aim of this study was to assess the incidence, characteristics, treatment, and outcomes in relation to age among patients with AMI complicated by CS (AMI-CS) compared with those without CS, in a nationwide cohort between 2013 and 2022.

Methods

Data sources

The Norwegian Myocardial Infarction Registry (NORMI) serves as a national quality register of all patients admitted to hospitals in Norway with AMI. Registration is mandatory for all hospitals according to the Norwegian Cardiovascular Disease Registry Regulation and the Norwegian Health Register Act. NORMI collects comprehensive data, including demographic information, cardiovascular risk factors, medical history, medications, presenting symptoms, clinical findings, electrocardiogram (ECG) data, in-hospital interventions and complications, and patient outcomes. Deaths after discharge are automatically updated in NORMI from the National Population Register. The registration process and data quality in NORMI have been described previously.^{17,18}

Additionally, information on causes of deaths was obtained retrospectively by linking NORMI to the Norwegian Cause of Death Registry, which provides causes of all deaths in Norway (<https://helsedata.no/en/forvaltere/norwegian-institute-of-public-health/norwegian-cause-of-death-registry/>).

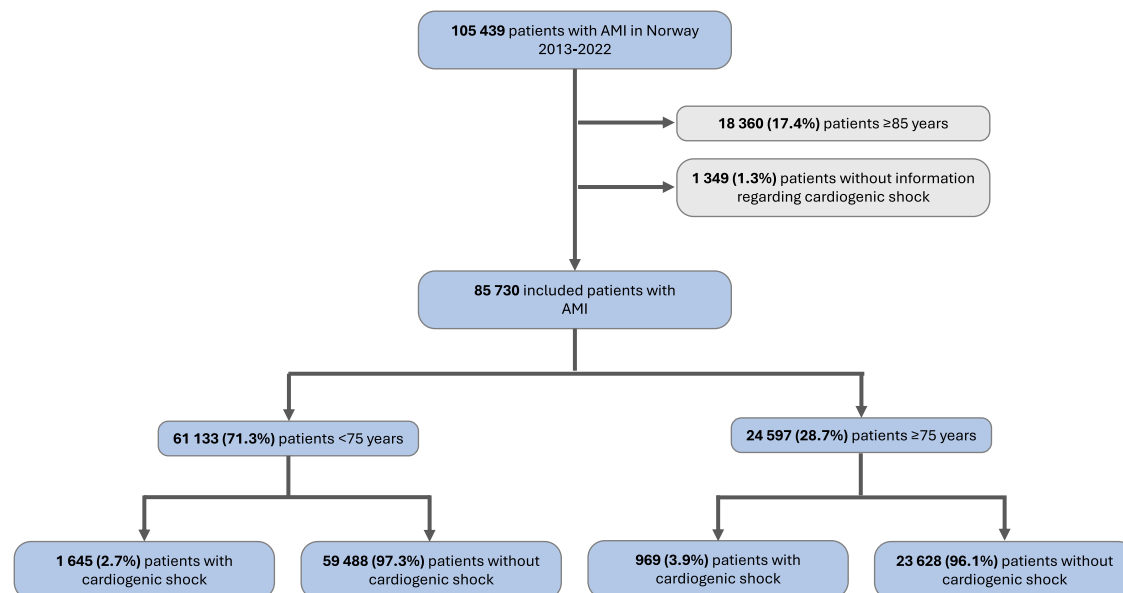


Figure 1 Cohort creation flow chart. AMI, acute myocardial infarction.

Study population

This cohort study included all patients in Norway hospitalized with AMI between 1 January 2013, and 31 December 2022, registered in the NORMI. For patients who experienced multiple AMIs during the study period, only data from the first event (index AMI) were used for analysis. We excluded patients who were 85 years or older for the following reasons: (i) presumed high rates of treatment restrictions related to factors that cannot be easily identified or adjusted for based on variables from a registry, (ii) high probability of competing risk of death unrelated to CS, and (iii) a low number of patients. We also excluded AMI patients with missing information on CS status. Furthermore, all patients were grouped according to age <75 years and 75–84 years. The age threshold was chosen based on the results from the original SHOCK trial.⁷

Definitions

The diagnosis and classification of AMI was determined by the treating physician according to the diagnostic criteria outlined in the Third (2013–18) and Fourth (2019–22) Universal Definition of Myocardial Infarction.^{19,20} The classification was based on a combination of clinical information, ECGs, biochemical markers, coronary angiography, and non-invasive imaging.

AMI-related CS was defined by NORMI, in alignment with ESC guidelines, as signs and symptoms of tissue hypoperfusion caused by AMI. Clinical criteria included a systolic blood pressure of <90 mm Hg for ≥30 min, or the requirement for pharmacological or mechanical support to maintain systolic blood pressure ≥90 mm Hg, along with symptoms of tissue hypoperfusion (e.g. anxiety, altered consciousness, cool extremities, increased lactate, and/or reduced diuresis).⁶

Outcomes and follow-up

The primary outcomes were in-hospital and long-term all-cause mortality. Follow-up data were available until 31 December 2022. The causes of death were obtained from the Norwegian Cause of Death registry, defining ischaemic heart disease as death caused by any of the International Classification of Diseases 10th revision (ICD-10) codes I20–I25.

Patient and public involvement

This study used existing data from mandatory Norwegian national health registries and consent by the patient was not required.

Statistics

Continuous variables are presented as mean ± standard deviation (SD) or median (25th percentile and 75th percentiles), and differences between groups were analysed using independent samples t-tests or Mann–Whitney U-tests, as appropriate. Categorical variables are reported as numbers and percentages (of non-missing values if missing >2%), with differences between groups assessed using chi-squared tests. Kaplan–Meier survival curves were generated, and differences between groups were evaluated using the Log-rank test. The Life Table method was used to estimate cumulative mortality. Logistic regression was used to calculate multivariable-adjusted odds ratios (ORs), and cox proportional hazards regression was used to calculate multivariable-adjusted hazard ratios (HRs) with 95% confidence intervals (CIs), adjusting for age, sex, smoking, previous AMI, stroke, diabetes, antihypertensive therapy, prior heart failure, renal failure [estimated glomerular filtration rate (eGFR) < 60 mL/min/1.73m²] and out-of-hospital cardiac arrest (OHCA). The covariates were selected *a priori* based on existing knowledge of potential confounders for the association between CS and mortality. Proportional hazard assumptions were assessed using Schoenfeld residuals and log-log survival plots. Time trends were analysed using a log-linear model by Joinpoint Regression Program (version 4.0; SEER software, National Cancer Institute, USA) and are presented as the expected annual percent changes (APC) with 95% CI. Other analyses were performed using STATA, version 18 (StataCorp LLC, College Station, TX, USA). A *P*-value of <0.05 was considered statistically significant.

Ethics

The study was conducted in accordance with the Declaration of Helsinki. The study was approved by the Regional Committee for Medical and Health Research Ethics North (REK 2016/170).

Results

Study population

A total of 105 439 patients with AMI were registered in the NORMI between 1 January 2013, and 31 December 2022 (Figure 1). After excluding patients ≥85 years and patients with missing CS data, 85 730 (98.5%) patients were included in the analysis. Of these, 25 661 (39.9%) were women, and 24 597 (28.7%) were 75–84 years old. We

identified a total of 1645 (2.7%) patients <75 years of age and 969 (3.9%) patients 75–84 years of age with AMI complicated by CS during these 10 years in Norway. No significant temporal trends were observed for the proportions of CS among AMI patients <75 years [change per year: -0.5 (95% CI: -4.7 to 3.7)] or ≥ 75 years [change per year: -2.3% (95% CI: -6.1 to 1.5)] during the study period (Figure 2).

The clinical characteristics of AMI patients with and without CS in relation to age are presented in Table 1. Although women were less likely to be admitted with AMI than men (29.9% vs. 70.1%) and the relative proportion of women increased with age, no significant gender differences were observed in the occurrence of CS for any of the age groups. A total of 417 out of 15 471 (2.7%) women <75 years with AMI and 393 out of 10 190 (3.9%) women 75–84 years with AMI experienced CS, compared with 1228 out of 45 662 (2.7%) and 576 out of 14 407 (4.0%) men, respectively.

Regardless of age, patients with AMI complicated by CS were more likely to be smokers and to have a history of heart failure (9% vs. 5%), diabetes (23% vs. 20%), or renal failure (47% vs. 21%) compared with those without CS, but less likely to have been treated with percutaneous coronary intervention (PCI) previously (14% vs. 17%). As expected, older patients had higher burden of comorbidities than younger patients. Similar patterns were observed when the <75-year age group was further divided into <60 years and 60–74 years (see Supplementary material online, Table S1). The time from symptom onset to hospital admission was significantly shorter in AMI-CS patients compared with those without CS. Approximately one in three AMI-CS patients <75 years of age, and one in eight AMI-CS patients 75–84 years of age, were admitted following an OHCA.

Assessment and treatment

Information about assessment and treatment related to age are summarized in Table 2.

AMI-patients complicated by CS were more likely to be diagnosed with ST-elevation myocardial infarction (STEMI) compared with those without CS, irrespective of age groups (Figure 2). A total of 5.2% STEMI cases <75 years of age and 9.8% STEMI cases 75–84 years of age were complicated by CS compared with 1.4% ($P < 0.001$) and 2.4% ($P < 0.001$) non-STEMI cases, respectively. Patients with AMI-CS in both age groups had higher maximum cardiac troponin level compared with those without CS. They were also more likely to have left ventricular ejection fraction (LVEF) $\leq 40\%$.

AMI-CS patients were less likely to undergo coronary angiography and PCI compared with AMI patients without CS. The rate of coronary angiography in CS patients increased from 72% (2014) to 87% (2021) in patients <75 years of age [APC 1.8 (95% CI: 0.8–2.8)], and from 38% (2013) to 70% (2022) in patients 75–84 years of age [APC 7.7 (95% CI: 3.1–12.5)]. No sex differences in the rate of coronary angiography among patients with AMI complicated by CS were observed.

Cardiopulmonary resuscitation out-of-hospital or in-hospital were performed in approximately half of the patients with AMI-CS, and one out of three AMI-CS patients were treated with MCS. Only 15% of AMI-CS patients 75–84 years of age were treated with MCS compared with 43% of patients <75 years of age. No temporal trends were observed in the use of MCS in Norway during the study period (<75 years: APC 2.2 [95% CI: (-0.4) –4.8]; 75–84 years: APC 8.6 [95% CI: (-4.3) –23.2]). Women with AMI-CS were less likely to be treated with MCS compared with men (<75 years: adjusted OR 0.70, (95% CI: 0.58–0.86), $P < 0.001$, 75–84 years: adjusted OR 0.58, (95% CI: 0.39–0.87), $P = 0.008$).

In-hospital mortality

A total of 782 (47.5%) AMI-CS patients <75 years of age died during the index hospitalization compared with 1022 (1.7%) patients without CS. In patients 75–84 years of age, 692 (71.4%) AMI-CS patients died

during the index hospitalization vs. 1355 (5.7%) patients without CS. Adjusted ORs for in-hospital mortality in AMI patients with CS compared with patients without CS were 30.1 (95% CI: 26.4–34.3), and 38.1 (95% CI: 32.6–44.7) in patients <75 and 75–85 years of age, respectively. The adjusted OR for in-hospital mortality for AMI-CS patients 75–84 years compared with patients <75 years of age was 2.48 (95% CI 2.06–2.99). No sex differences were observed for in-hospital mortality among patients with AMI complicated by CS during the study period.

A substantial proportion of patients with CS was admitted after OHCA, and these patients had significantly higher in-hospital mortality compared with CS patients without OHCA (59.6% vs. 40.4%, $P < 0.001$). In patients with OHCA, adjusted ORs for in-hospital mortality in AMI-CS compared with AMI without CS were 51.8 (95% CI: 44.6–60.1) for patients <75, and 42.0 (95% CI: 35.7–49.5) for patients 75–84 years. In patients with OHCA, adjusted ORs for in-hospital mortality AMI-CS compared with AMI without CS were 6.7 (95% CI: 5.4–8.4) and 11.6 (95% CI: 7.0–19.2) for patients <75 and 75–84 years, respectively.

Among AMI-CS patients treated with MCS, in-hospital mortality rates were 41.7% for those aged <75 years and 58.6% for those aged 75–84 years.

Ischaemic heart disease was reported as the primary cause of death in 83.2% of the AMI-CS patients who died during the index hospitalization.

Treatment at hospital discharge

A total of 81 879 (95.5%) AMI patients were discharged alive. The prescription of cardiovascular drugs at hospital discharge is presented in Table 3. Patients with AMI-CS were more often prescribed heart failure medications and oral anticoagulation, and less likely to be prescribed lipid lowering therapy, compared with patients without CS.

Long-term mortality

The median follow-up period was 3.9 (25th, 75th percentiles; 1.5, 6.8) years. Over the total follow-up period, 61.0% ($n = 1004$) of AMI-CS patients <75 years and 88.0% ($n = 853$) of the AMI-CS patients 75–84 years died, compared with 13.3% ($n = 8478$) and 46.3% ($n = 10 931$) of those without CS. Figure 3 presents survival curves for AMI patients with and without CS for patients <75 years and patients 75–84 years of age.

The estimated cumulative mortality rates for AMI patients with and without CS are described in Table 4. When considering age on a continuous scale, a non-linear association ($P < 0.001$) between age and mortality in patients with AMI-related CS was clearly demonstrated (Figure 4).

Patients with AMI-CS who survived to hospital discharge remained at a higher long-term mortality risk compared with AMI patients without CS. In patients <75 years: of age, the adjusted HR was 1.91 (95% CI 1.67–2.18, $P < 0.001$), and in patients 75–84 years, the adjusted HR was 1.65 (95% CI: 1.41–1.93, $P < 0.001$) (P -for interaction = 0.002 for <75 vs. 75–84 years) (Figure 5). Long-term mortality after discharge did not differ between women and men with AMI-CS [adjusted HR 1.01 (95% CI 0.98–1.05), $P = 0.38$].

In AMI-CS patients who survived to hospital discharge, cardiac arrest had no impact on long-term survival [adjusted HR 0.88 (95% CI: 0.71–1.10)]. In contrast, in AMI-CS patients who survived to hospital discharge, patients with LVEF $\leq 40\%$ had worse outcomes compared with patients with LVEF $> 40\%$ [adjusted HR 1.63 (95% CI: 1.31–2.03)].

Among patients who died after hospital discharge, ischaemic heart disease (ICD-10 codes I20–I25) was the reported cause of death in 22.2 and 21.4% of the patients <75 and 75–84 years without CS, and in 42.3 and 41.0% of the patients <75 and 75–84 years with AMI-CS.

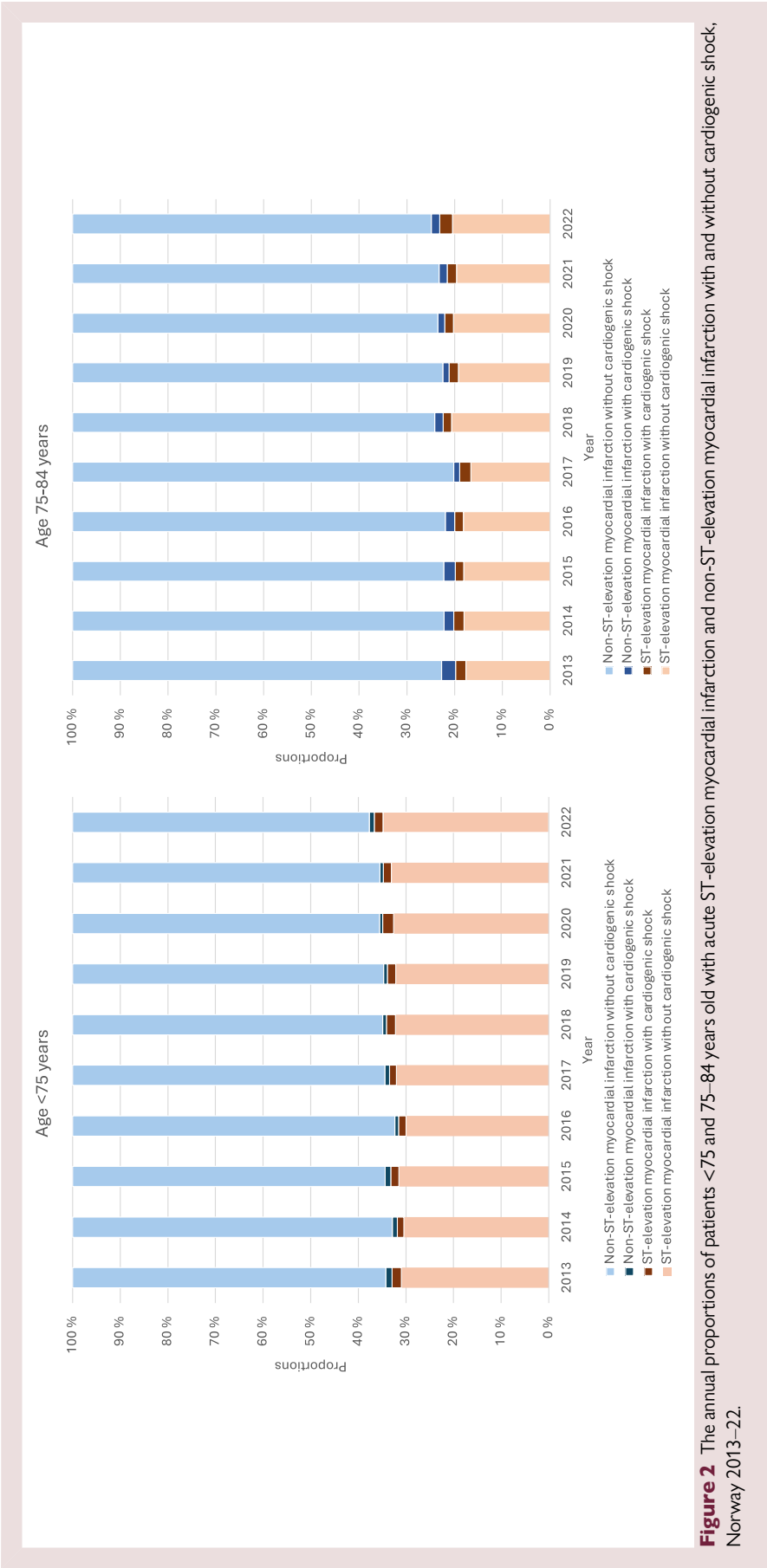


Figure 2 The annual proportions of patients <75 and 75–84 years old with acute ST-elevation myocardial infarction with and without cardiogenic shock, Norway 2013–22.

Table 1 Clinical characteristics of patients with acute myocardial infarction with and without cardiogenic shock according to age, Norway 2013–22

	Age <75 years		<i>P</i> ¹	Age 75–84 years		<i>P</i> ²	<i>P</i> ³
	Cardiogenic shock	No cardiogenic shock		Cardiogenic shock	No cardiogenic shock		
	<i>n</i> = 1645	<i>n</i> = 59488		<i>n</i> = 969	<i>n</i> = 23628		
Women (<i>n</i>)	417 (25.3%)	15054 (25.3%)	0.97	393 (40.6%)	9797 (41.5%)	0.57	<0.001
Mean age [year (SD)]	62.5 ± 9.0	61.3 ± 9.2	<0.001	79.4 ± 2.9	79.4 ± 2.9	0.73	NA
Median age [year (IQR)]	64 (57, 70)	63 (55, 69)	<0.001	79 (77, 82)	79 (77, 82)	0.74	NA
Median body mass index (SD)	27.2 ± 5.0	27.9 ± 4.8	<0.001	25.3 ± 4.3	25.9 ± 4.4	0.003	<0.001
Smoking (<i>n</i>) (missing 7.2%)	671/1318 (50.9%)	22217/56465 (39.4%)	<0.001	177/746 (23.7%)	3682/21005 (17.5%)	<0.001	<0.001
Previous myocardial infarction (<i>n</i>)	285 (17.3%)	9111 (15.3%)	0.026	239 (24.7%)	5883 (24.9%)	0.87	<0.001
Previous percutaneous coronary intervention (<i>n</i>)	217 (13.2%)	9083 (15.3%)	0.021	150 (15.5%)	4786 (20.3%)	<0.001	0.104
Previous coronary artery bypass grafting (<i>n</i>)	109 (6.6%)	3371 (5.7%)	0.10	116 (12.0%)	3148 (13.3%)	0.22	<0.001
Previous heart failure diagnosis (<i>n</i>)	124 (7.5%)	2008 (3.4%)	<0.001	123 (12.7%)	2287 (9.7%)	0.002	<0.001
Previous stroke (<i>n</i>)	121 (7.4%)	2858 (4.8%)	<0.001	128 (13.2%)	2810 (11.9%)	0.22	<0.001
Diabetes (<i>n</i>)	360 (21.9%)	11072 (18.6%)	<0.001	254 (26.2%)	5348 (22.6%)	0.009	0.012
Hypertension (<i>n</i>)	668 (40.6%)	24402 (41.0%)	0.74	556 (57.4%)	13528 (57.3%)	0.94	<0.001
Renal failure (eGFR < 60 mL/min/1.73m ²) (<i>n</i>) (missing 2.8%)	625/1498 (41.7%)	7257/5879 (12.5%)	<0.001	615/889 (69.2%)	9990/23079 (43.3%)	<0.001	<0.001
Medication prior to index myocardial infarction							
Acetylsalicylic acid (<i>n</i>)	405 (24.6%)	16594 (27.9%)	0.003	363 (37.5%)	10304 (43.6%)	<0.001	<0.001
Anticoagulation therapy (<i>n</i>)	126 (7.7%)	3682 (6.2%)	0.015	168 (17.3%)	4332 (18.3%)	0.43	<0.001
Angiotensin-converting enzyme (ACE) inhibitors/angiotensin receptor blockers (ARB) (<i>n</i>)	478 (29.1%)	18891 (31.8%)	0.020	416 (42.9%)	10243 (43.4%)	0.80	<0.001
Beta blocker (<i>n</i>)	388 (23.6%)	13464 (22.6%)	0.36	415 (42.8%)	9630 (40.8%)	0.20	<0.001
Diuretics (<i>n</i>)	260 (15.8%)	8631 (14.5%)	0.14	296 (30.5%)	6691 (28.3%)	0.13	<0.001
Lipid lowering therapy (<i>n</i>)	465 (28.3%)	19064 (32.0%)	<0.001	377 (38.9%)	10596 (44.8%)	<0.001	<0.001
Out-of-hospital cardiac arrest (<i>n</i>)	487 (29.6%)	2106 (3.5%)	<0.001	126 (13.0%)	538 (2.3%)	<0.001	<0.001
Median delay from debut of symptom to hospitalization (min, IQR)	145 (77, 397)	200 (108, 559)	<0.001	195 (94, 629)	250 (123, 700)	<0.001	<0.001

SD, standard deviation; IQR, interquartile range; eGFR, estimated glomerular filtration rate.

*P*¹ and *P*²: No cardiogenic shock vs. cardiogenic shock.*P*³: Cardiogenic shock <75 years vs. cardiogenic shock 75–84 years

Discussion

In this nationwide cohort study from 2013 to 2022 of 86 730 AMI patients, a total of 2614 (3%) patients developed CS. The proportion of AMI patients who developed CS was higher in AMI patients 75–84 years of age (3.9%) compared with AMI patients <75 years of age (2.7%). No temporal changes in the proportion of AMI patients who developed CS were observed. The in-hospital mortality rates of patients with AMI-CS were high, particularly for patients 75–84 years of age, where more than two out of three patients died in-hospital. Patients who survived until hospital discharge, had only moderately elevated risk of long-term mortality compared with non-CS patients and the increased risk was comparable between patients <75 years and 75–84 years of age.

The main strength of this study is the extensive and unselected nationwide cohort, which includes nearly all patients <85 years hospitalized with AMI in Norway from 2013 to 2022, along with nearly complete follow-up data. Nevertheless, several important limitations need to be mentioned. The NORMI exclusively captured AMI patients who were hospitalized. Adjustments for covariates were limited to those recorded in the NORMI, leaving unmeasured confounders present. Furthermore, the registration of CS within the NORMI relied on clinical assessment conducted during the index hospitalization for AMI, which may have resulted in inaccuracies in the definition of CS. The use of vasoactive agents, including vasopressors and inotropes, was not documented in the NORMI. Additionally, only a single blood pressure measurement taken immediately upon admission is recorded

Table 2 Assessment and treatment of patients with acute myocardial infarction with and without cardiogenic shock according to age, Norway 2013–22

	Age <75 years			Age 75–84 years			<i>P</i> ³
	Cardiogenic shock	No cardiogenic shock	<i>P</i> ¹	Cardiogenic shock	No cardiogenic shock	<i>P</i> ²	
	<i>n</i> = 1645	<i>n</i> = 59488		<i>n</i> = 969	<i>n</i> = 23628		
ST-elevation myocardial infarction (STEMI) (<i>n</i>) (missing 4.0%)	1071/1487 (72.0%)	19466/57115 (34.1%)	<0.001	500/896 (55.8%)	4626/22766 (20.3%)	<0.001	<0.001
Prehospital fibrinolytic therapy (<i>n</i>)	114 (6.9%)	1889 (3.2%)	<0.001	34 (3.5%)	325 (1.4%)	<0.001	<0.001
In-hospital fibrinolytic therapy (<i>n</i>)	102 (6.2%)	1031 (1.7%)	<0.001	40 (4.1%)	209 (0.9%)	<0.001	<0.001
Coronary angiography (<i>n</i>)	1339 (81.4%)	52039 (87.5%)	<0.001	492 (50.8%)	14940 (63.2%)	<0.001	<0.001
Percutaneous coronary intervention (PCI) (<i>n</i>)	1174 (71.4%)	40314 (67.8%)	0.002	398 (41.1%)	10639 (45.0%)	0.015	<0.001
Median delay from symptom onset to PCI (hours, IQR) in patients with STEMI	3.8 (2.2, 8.5)	3.7 (2.3, 7.9)	0.89	4.2 (2.5, 9.3)	4.4 (2.6, 10.4)	0.26	0.160
Mean systolic blood pressure [mmHg, (SD)] at hospital admission	107.7 ± 35.0	142.4 ± 26.8	<0.001	109.8 ± 34.2	142.6 ± 29.5	<0.001	0.159
Mean diastolic blood pressure [mmHg, (SD)] at hospital admission	67.5 ± 23.5	83.6 ± 16.0	<0.001	65.3 ± 20.5	77.7 ± 16.4	<0.001	0.025
Left ventricular ejection fraction (LVEF) ≤ 40% during index hospitalization (<i>n</i>) (missing 22.9%)	806/1361 (59.2%)	4601/47479 (9.7%)	<0.001	477/710 (67.2%)	2978/16568 (18.0%)	<0.001	<0.001
Peak cardiac troponin T concentration (median) (ng/L, IQR) (missing 17.7%)	5334 (1489, 12046)	674 (151, 2647)	<0.001	3136 (701, 9616)	503 (140, 1953)	<0.001	<0.001
In-hospital cardiopulmonary resuscitation (<i>n</i>)	768 (46.7%)	1319 (2.2%)	<0.001	312 (32.2%)	576 (2.4%)	<0.001	<0.001
Mechanical circulatory support (<i>n</i>) ^a	696 (42.3%)	350 (0.6%)	<0.001	145 (15.0%)	93 (0.4%)	<0.001	<0.001
Respiratory support (<i>n</i>) ^b	1095 (66.6%)	2984 (5.0%)	<0.001	464 (47.9%)	1762 (7.5%)	<0.001	<0.001

SD, standard deviation; IQR, interquartile range.

*P*¹ and *P*²: No cardiogenic shock vs. cardiogenic shock.*P*³: Cardiogenic shock <75 years vs. cardiogenic shock 75–84 years.^aVeno-arterial extracorporeal life-support (ECLS), intra-aortic balloon pump (IABP), percutaneous microaxial pumps, or paracorporeal centrifugal pumps.^bNon-invasive respiratory support or mechanical ventilation.

in NORMI, which may not reflect haemodynamic status in patients who developed CS later during hospitalization. The registry also did not differentiate between different MCS devices; however, intra-aortic balloon pump was the most used MCS in Norway in the study period. Moreover, NORMI did not include any information regarding decisions of withdrawal of life sustaining treatment (WLST). We utilized de-identified data from the NORMI and were unable to verify individual patient information through medical records. Patients with AMI enrolled early in the study period had a longer follow-up duration compared with those included later, potentially introducing follow-up bias in long-term outcome analyses. Additionally, we lacked information regarding post-discharge treatments and the quality of life of the survivors. Given the observational design of the study, it is not possible to establish causal relationships between treatment and clinical outcomes, necessitating careful interpretation of the findings.

The proportion of patients with CS in our cohort remained stable throughout the study period despite an annual decline in AMI cases recorded in the NORMI.^{10,11} Compared to previous reports, the incidence of CS among AMI patients in the present study was low.^{13,14,21} Notably, no significant gender differences were observed in the occurrence of CS, despite that the proportion of women presenting with AMI increased with advancing age.

Advanced age, the presence of multiple risk factors, a high burden of comorbidities, ST-elevation in ECG, low blood pressure, and high troponin concentration in plasma were associated with risk of developing CS. This aligns well with previously described predictors.²²

Early revascularization and restoration of coronary blood flow are critical for optimizing myocardial salvage and reducing mortality in patients with AMI.²³ Current guidelines from the ESC recommend timely coronary angiography for most AMI patients including patients with CS.²⁴ Few trials have directly compared the effectiveness of early revascularization in AMI-CS patients.^{25–27} In the SHOCK trial, immediate revascularization with PCI or coronary artery bypass graft surgery did not reduce 30-day mortality in patients with STEMI and CS, but 6-month survival was improved compared with patients assigned to medical therapy (50% vs. 63%).^{7,28,29} However, the mean age was 66 (±10) years in the SHOCK-trial. In patients ≥75 years, both 30-day and 6-month mortality were higher in the group of re-vascularized patients compared with patients treated with medical therapy. In clinical practice, these results were followed by a more conservative strategy in elderly AMI patients with CS. Later results from the SHOCK Trial Registry, however, showed lower mortality rates in elderly patients selected for early revascularization compared with those receiving no or later revascularization procedure.²⁷ Reduced 1-year mortality in patients

Table 3 Medication at hospital discharge in patients discharged alive after myocardial infarction with and without cardiogenic shock according to age, Norway 2013–22

	Age <75 years		<i>P</i> ¹	Age 75–84 years		<i>P</i> ²	<i>P</i> ³
	Cardiogenic shock	No cardiogenic shock		Cardiogenic shock	No cardiogenic shock		
	<i>n</i> = 863	<i>n</i> = 58466		<i>n</i> = 277	<i>n</i> = 22273		
Acetylsalicylic acid (<i>n</i>) (missing 12.3%)	617/679 (90.9%)	48196/51119 (94.3%)	<0.001	181/221 (81.9%)	16734/19790 (84.6%)	0.274	<0.001
Dual antiplatelet therapy (<i>n</i>) (missing 12.4%)	560/678 (82.6%)	43371/51075 (84.9%)	0.094	148/221 (67.0%)	13330/19750 (67.5%)	0.87	<0.001
Anticoagulation therapy (<i>n</i>) (missing 12.5%)	178/675 (26.4%)	6666/51023 (13.1%)	<0.001	79/221 (35.8%)	5493/19705 (27.9%)	0.010	0.007
Angiotensin-converting enzyme (ACE) inhibitors/angiotensin receptor blockers (ARB) (<i>n</i>) (missing 12.3%)	518/679 [76.3%]	29177/51112 (57.1%)	<0.001	160/220 (72.7%)	11662/19763 (59.0%)	<0.001	0.286
Beta-blocker (<i>n</i>) (missing 12.3%)	550/678 (81.1%)	34753/51107 (68.0%)	<0.001	177/221 (80.1%)	14034/19768 (71.0%)	0.003	0.735
Diuretics (<i>n</i>) (missing 12.4%)	327/677 (48.3%)	9792/51107 (19.2%)	<0.001	149/221 (67.4%)	7475/19765 (37.8%)	<0.001	<0.001
Lipid lowering therapy (<i>n</i>) (missing 12.3%)	572/677 (84.5%)	47393/51105 (92.7%)	<0.001	174/221 (78.7%)	16435/19769 (83.1%)	0.083	0.048

*P*¹ and *P*²: No cardiogenic shock vs. cardiogenic shock.
*P*³: Cardiogenic shock <75 years vs. cardiogenic shock 75–84 years.

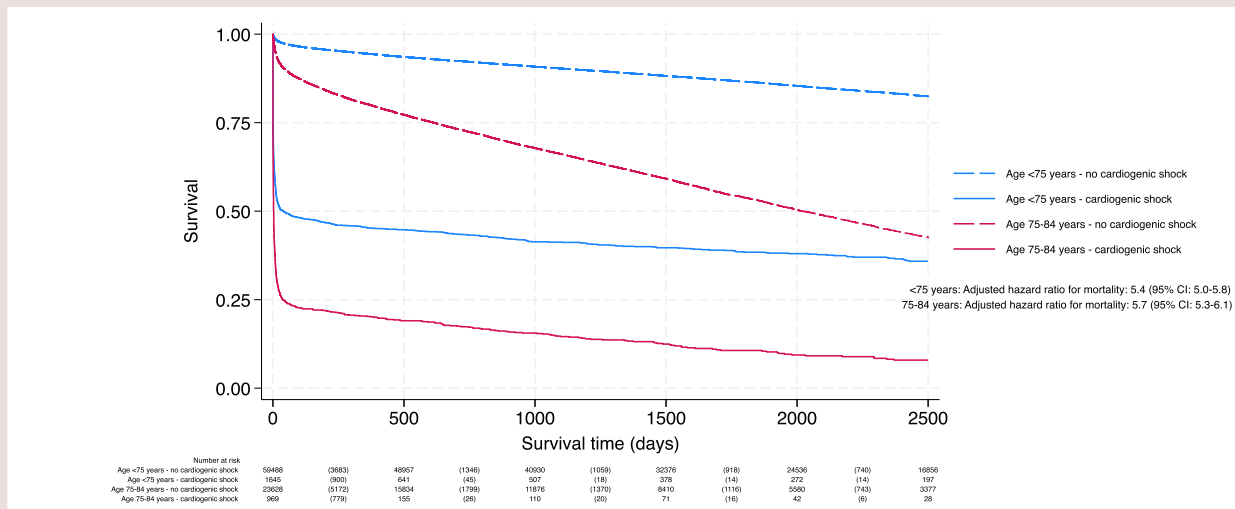


Figure 3 Kaplan–Meier curves of survival in patients with acute myocardial infarction with and without cardiogenic shock in relation to age, Norway, 2013–22.

with STEMI, CS, and successful PCI was also reported from the Bremen STEMI registry who included approximately 28% patients >75 years of age.³⁰ Still, the impact of early revascularization in elderly people with AMI-CS has not been settled in a randomized controlled trial. The proportion undergoing coronary angiography and PCI in this study was lower than might be expected, particularly in patients 75–84 years. However, it is noteworthy that these proportions increased during the study period. The lower than expected proportions may partially be explained by the high incidence of cardiac arrest with high early mortality, which could have limited the opportunity for early invasive evaluation and treatment. Furthermore, increasing comorbidity with increasing age, and consequently a higher risk of complications, might have influenced the choice of treatment strategy in the elderly patient group. Patients who were treated with PCI, were more likely to survive

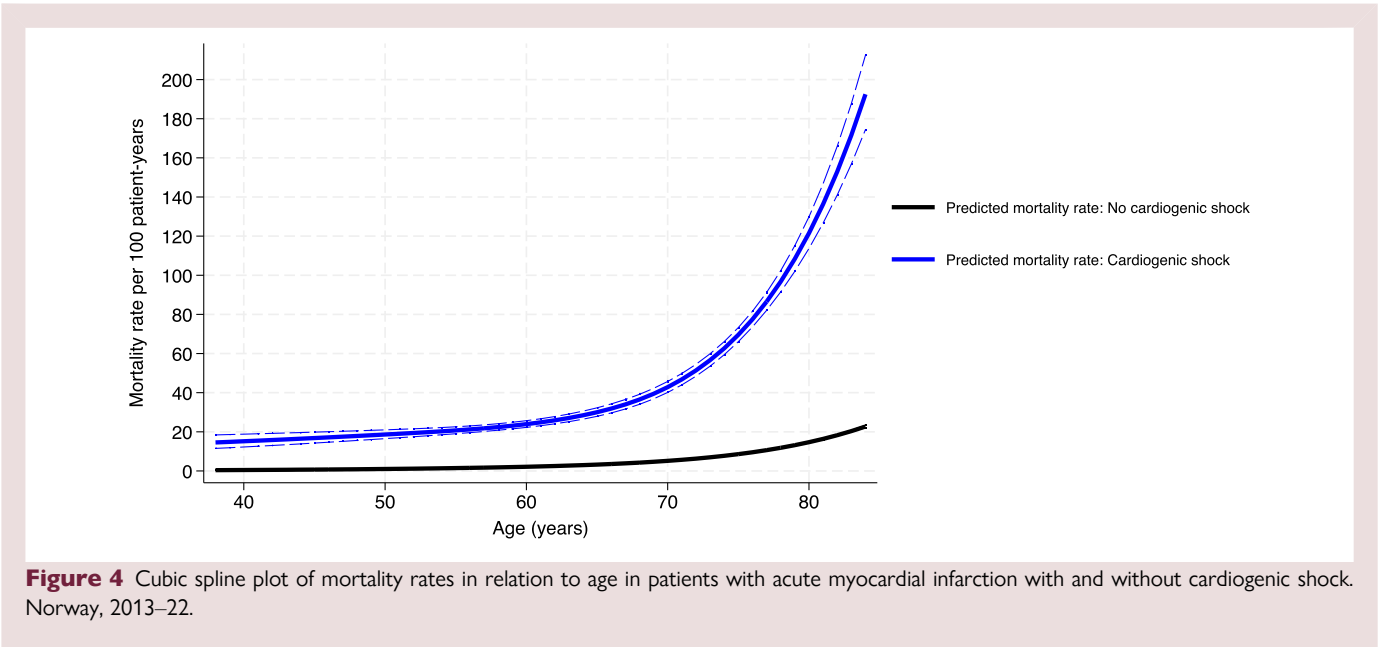
to discharge. However, due to the high risk of early death (before angiography could be performed) and several selection biases, the impact of PCI on in-hospital mortality cannot be conclusively determined in this observational register study.

There are no conclusive studies evaluating the relative efficacy of different MCS devices in the setting of AMI-CS in any age groups.^{31–35} The DanGer Shock trial recently reported improved survival after STEMI-related CS with use of a micro-axial flow pump, but patients aged ≥77 years did not attain any mortality benefit of such treatment.^{36,37} In our cohort, a total of 42% of the patients <75 years and 15% of the patients 75–84 years with AMI-related CS received MCS therapy, and both the in-hospital and long-term mortality were high in these patients. Women were less likely to receive treatment with MCS compared with men, however, the survival rates were

Table 4 Estimated cumulative mortality (life table method) in patients with acute myocardial infarction with and without cardiogenic shock in Norway 2013–2022

	Age <75 years		Age 75–84 years	
	Cardiogenic shock	No cardiogenic shock	Cardiogenic shock	No cardiogenic shock
30-Day mortality [%; (CI)]	49.4 (47.0–51.9)	2.4 (2.2–2.5)	73.9 (71.1–76.6)	8.8 (8.4–9.1)
1-Year mortality [%; (CI)]	54.7 (52.3–57.1)	5.6 (5.4–5.8)	79.8 (77.2–82.3)	19.9 (19.4–20.4)
1-Year mortality, discharged alive [%; (CI)]	13.3 (11.2–15.8)	3.9 (3.7–4.1)	29.0 (23.9–34.8)	15.0 (14.5–15.5)
2-Year mortality [%; (CI)]	56.4 (54.1–59.9)	7.7 (7.5–7.9)	82.7 (80.2–85.0)	27.2 (26.7–27.8)
2-Year mortality, discharged alive [%; (CI)]	16.8 (14.4–19.6)	6.1 (5.9–6.3)	39.0 (33.3–45.4)	22.8 (22.2–23.4)

CI, confidence interval.



similar. Sex differences in the use of MCS have been reported in prior studies.³⁸ Such differences are difficult to explain, but may be partially attributable to age differences as well as comorbid conditions in women that could not be accounted for in this registry-based analysis. It is not possible to draw any conclusions about the impact of MCS on outcome in the elderly from this observational study. Routine use of MCS is not recommended for CS complicating AMI in current ESC guidelines.²⁴ MCS in elderly patients is extremely resource consuming and initiation of such therapy requires careful ethical considerations. However, 41% of the AMI-CS patients 75–84 years treated with MCS survived to hospital discharge suggesting a reasonable selection of elderly patients for such advanced treatment.

The short-term mortality rates for patients with AMI-CS were high in our study, especially in the elderly (71.4% in-hospital mortality). High short-term mortality in patients with CS related to acute coronary syndrome/AMI has also been shown in studies from Denmark and Sweden in 1995–2017.^{13–15} A recent systematic review and meta-analysis of all-cause CS by Jung et al., including 160 studies and 2 459 703 patients, reported a median in-hospital mortality of 41%.² In CS patients >75 years of age, the short-term-mortality has previously been reported to be approximately 75%.²¹ The recent

findings from the randomized clinical DanGer Shock trial also align with our observations regarding the poor prognosis of older patients with AMI-CS.³⁷ Despite significant advances in the treatment of AMI, including the implementation of timely reperfusion therapies, the short-term mortality associated with CS in elderly patients has not improved substantially over the past decades.^{2,21,39,40} The high short-term mortality underscores the severity of CS as a complication of AMI, emphasizing the need for enhanced prevention strategies, early identification, and improved treatment.

The in-hospital mortality in AMI patients complicated by OHCA and CS was high. However, among those who were discharged alive, long-term mortality rates were comparable between patients with and without cardiac arrest. Notably, we have recently reported similar findings in the broader population of AMI patients with OHCA, further supporting this observation.⁴ The occurrence of OHCA with suspicion of serious brain injury may have influenced a decision of WLST, especially in older patients, where prognostic concerns could impact the intensity of interventions.

The long-term post-discharge mortality following AMI complicated by CS is less studied than the short-term mortality. In this study, the 1-year post-discharge mortality rates in AMI-CS patients <75 years

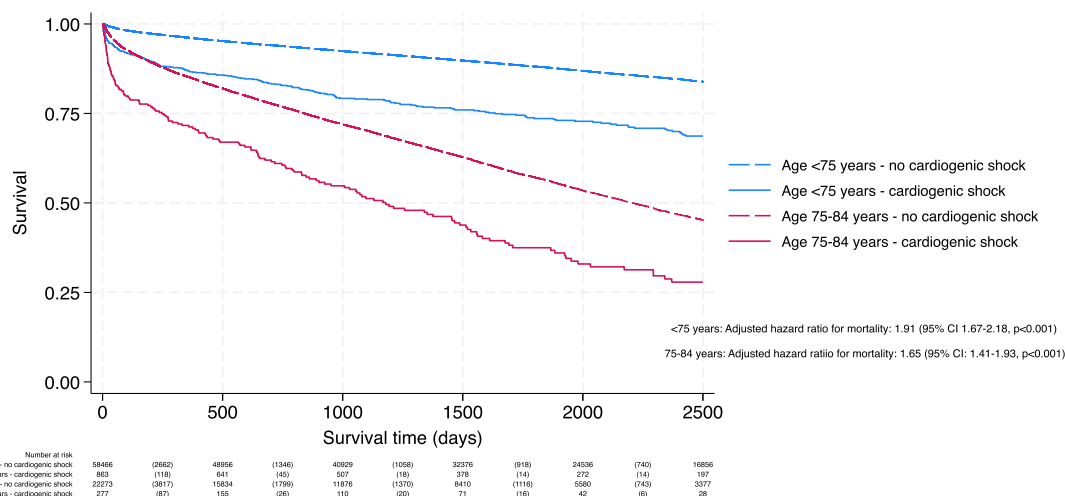


Figure 5 Kaplan–Meier curves of survival in patients with acute myocardial infarction with and without cardiogenic shock surviving until hospital discharge (landmark analysis).

and 75–84 years of age surviving to hospital discharge were 13 and 29%, respectively. Patients with AMI-CS discharged alive in both age groups exhibiting only a moderately elevated mortality risk compared with those without CS (<75 years: adjusted HR 1.91 (95% CI 1.67–2.18), 75–84 years: adjusted HR 1.65 (95% CI: 1.41–1.93)). Furthermore, the elevated risk of mortality in AMI-CS survivors are reported to decline with time from index event.⁴¹ A high rate of early revascularization and high prescription rates of dual antiplatelet therapy, heart failure medication, and beta-blockers in CS survivors may contribute to the relatively high survival rates in Norway.

In conclusion, CS continues to pose a significant challenge, with severe consequences for both short- and long-term survival after AMI. The in-hospital mortality rates for AMI-CS patients <75 years as well as for patients 74–85 years of age were high: almost one out of two patients <75 years, and more than two out of three patients 75–84 years of age, died in-hospital. However, patients who survived until hospital discharge exhibiting only a moderately elevated mortality risk compared with AMI patients without CS. The present study underscores the necessity of including older patients in randomized clinical trials to better define optimal management strategies for AMI complicated by CS, particularly in the context of an ageing population.

Supplementary material

Supplementary material is available at *European Heart Journal: Acute Cardiovascular Care* online.

Author contributions

All authors were responsible for the conception of the study, and the analysis and interpretation of data. J.J. drafted the manuscript, P.L.M., G.Ø.A., and S.H. critically revised the manuscript. All gave final approval and agreed to be accountable for all aspects of work ensuring its integrity and accuracy.

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Conflict of interest: J.J. has received speaking fees from Amgen, AstraZeneca, Bristol Myers Squibb, Boehringer Ingelheim, Novartis, Novo Nordisk, Pfizer, and Sanofi. P.L.M. has consulted for and/or received speaker fees from Amgen, Amarin, AstraZeneca, Bayer, Bristol Myers Squibb, Boehringer Ingelheim, Novartis, Novo Nordisk, Orion Pharma, Pharmacosmos, Sanofi, Us2.ai and Vifor. G.Ø.A. has received speaking fees from Orion Pharma. S.H. has received speaking fees from Boehringer Ingelheim, Bristol Myers Squibb, Pfizer and Sanofi.

Data availability

The data underlying this article were provided by the Norwegian Institute of Public Health under license/by permission. Data will be shared on request to the corresponding author with permission of the Norwegian Institute of Public Health.

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